

Antique Network QA - Remediation and Full Rerun

7 October 2026 | Patched local ParaBank | Fresh test evidence

Final full matrix: 67 Maven/TestNG executions; 67 passed; 0 failed; 0 errors; 0 skipped. Counts include both API seeds and count executions, not unique test designs.

Additional boundary regressions: 33/33 passed. Report-utility checks: exit 0, five checks. Original six-defect probes: 0/6 reproduced on the patched bank.

This result applies to the locally patched application. The original official demo and its earlier 10 failed cases per seed remain recorded. No public service was changed. Existing authorization, exact-money and deployed-schema assertions were retained.

Run	Tests	Pass	Fail	Error	Skip	Exit
ui-unit	6	6	0	0	0	0
ui-chrome-isolation	1	1	0	0	0	0
ui-chrome	7	7	0	0	0	0
ui-firefox-isolation	1	1	0	0	0	0
ui-firefox	7	7	0	0	0	0
api-unit	13	13	0	0	0	0
api-seed-42	15	15	0	0	0	0
api-seed-43	15	15	0	0	0	0
api-loans	1	1	0	0	0	0
api-reset	1	1	0	0	0	0

What failed originally and what changed

Original failing case	Cause found / local correction
crossCustomerTransactionRead	No account-owner enforcement. Shared servlet boundary checks the authenticated customer for account and transaction aliases across CXF, proxy and SOAP.
crossCustomerTransfer	Supplied foreign source IDs accepted. Ownership check precedes mutation; requester cannot debit an unrelated customer.
negativeDepositCannotReduceBalance	Negative amount passed to credit. External API and shared core reject negative/unsupported-precision amounts before mutation.
negativeWithdrawalCannotIncreaseBalance	Subtracting a negative amount created funds. Same validation prevents reversal of withdrawal direction.
invalidTransfers: insufficient	No funds predicate. Atomic JDBC debit requires balance >= requested debit; local no-overdraft policy is explicit.
invalidTransfers: negative	Negative transfer reversed movement. External API rejects it and shared core does not apply it.
invalidTransfers: zero	Original demo accepted zero and created ledger entries. Customer API rejects nonpositive amounts; internal zero loan down payments remain valid.
invalidTransfers: nonnumeric	Original response was 404 with empty body. Boundary validation returns 400 with meaningful plain-text content.
concurrentWithdrawals	Read-modify-write absolute balances could overwrite concurrent changes. JDBC arithmetic UPDATE plus sufficient-funds condition and existing transaction protect balance/ledger updates.
deployedXmlSchema	Domain roots/types used inconsistent namespaces. JAXB package namespace now matches the published service namespace while child fields stay unqualified; unchanged live-schema validation remains active.

Storefront synchronization correction

The first UI attempt during remediation had one Chrome requiredOptionsAndQuantity failure: the required-option error disappeared. Captured DOM retained has-error while text-danger was missing. OpenCart quantity blur starts a recurring-description AJAX call whose success removes text-danger. It could complete after cart validation. StorePage.add now blurs quantity and waits for that request before Add to Cart. The existing required-message and exact-total assertions remain intact.

Initial UI attempt: 22 checks, 21 passes, 1 failure, no errors/skips. Its logs, XML, screenshots and DOM are preserved under diagnostics/initial-ui. The table above is the separately recorded full rerun after the synchronization fix. It does not erase that initial failure or claim a repeated-run reliability target.

API session-client correction

The first Java attempt against the authenticated patched bank failed fixture setup: REST login returned 200, but own-account reads returned 401. BankClient.form replaced all saved cookies with only the latest response cookies; a successful registration response did not re-send Set-Cookie. The client now retains existing cookies and merges any rotated response cookies. No ownership assertion was weakened. The failed setup/skip results remain under diagnostics/initial-api; the table above records the complete run after correction. Live integration controls verify that legitimate authenticated operations succeed and foreign operations remain denied.

Namespace-aware parity correction

Once the live schema passed, jsonXmlParity exposed a framework bug: XML normalization used prefixed tag names as collection/field names. A qualified empty transactions element became a string instead of a list, and qualified numeric fields kept their prefixes and string values. Two new regression tests failed before the change (parity-regression-red.log). Normalization now uses DOM local names while retaining exact decimal comparison, canonical record IDs, missing-field checks and secure XML parsing. Live namespace/schema validation remains a separate unchanged assertion. The intermediate failure matrix remains in diagnostics/parity-intermediate-api.

Patch provenance and execution environment

QA repository base commit: 7672c3c46d849c8190121e92771217f91a2c0eaa. This delivery includes working-tree changes listed in source-file-checksums.json; the base commit alone does not identify the patched sources.

The Docker overlay derives from official ParaBank image sha256:cebe472d86a10be23ce448f0f419f3c939a818112ab78cd57fc09e185f35d1c0. Five upstream Java overlay files come from commit d9ff65e4447942c4e2b55cfa9a36ff478b032b80 with Apache 2.0 license retained. The deployed web.xml/cxf.xml were copied from that pinned image. New boundary/error/credential helpers are included. Docker compiles the overlay against actual pinned-image libraries using pinned Temurin 21 JDK; the QA framework targets Java 17 and runs under host JDK 24.

Patched image ID: sha256:af193d8c696082c5605fdb988be460dd3cf324d89c0b53926743d81906548a5

Windows 11; Maven Wrapper 3.9.9; Docker engine 29.8.2; Selenium 4.28.1; Chrome 154.0.8037.98 with two methods in parallel; Firefox 143.0.3 with one method thread. Each case uses separate admin/customer browser sessions. Serial Firefox is deliberate because prior parallel startup timed out on this host. Full Docker Grid and hosted CI were not run.

Final API runs target http://127.0.0.1:8081/parabank (loopback binding); UI targets http://localhost:8080/. Both are left running for review. Earlier probes used the equivalent localhost target; final boundary checks use 127.0.0.1. Fixtures are unique synthetic customers/products. Local loan/reset administration is authorized; no public mutations occurred in this remediation run.

Review findings and intermediate verification

Independent review confirmed that raw numeric-only path matching missed signed and percent-encoded IDs, SOAP metadata GET was parsed as an empty body, and optional REST/SOAP calls lacked outbound identity. These were corrected: URI decoding precedes integer ownership checks, WSDL metadata is accessible, and credentials propagate only to the same loopback application/port. Regression checks exercise those routes and legitimate controls.

The first compiled patch used an SQL predicate HSQLDB could not type-infer; a focused deposit failed with SQLSyntaxErrorException. The predicate now includes the sufficient-funds clause only for debits, comparing the column with a positive typed amount. Subsequent build/deployment logs and live regression evidence are retained. Intermediate failures are diagnostics, not counted as passing final tests.

An intermediate Java matrix after the cookie fix passed 14/15 API cases per seed. Root-only XML qualification left JAXB types in a different schema namespace; unchanged live-XSD validation detected the unresolved account type. The final package-level namespace aligns domain roots and types while preserving unqualified child fields. That intermediate matrix is retained in diagnostics/schema-intermediate-api.

boundary-before.json: 3/23 checks passed. Before/after probe inventories differ because review-driven alias and optional-mode cases were added.

boundary-after-precompat.json: 30/33 checks passed. Before/after probe inventories differ because review-driven alias and optional-mode cases were added.

boundary-after.json: 33/33 checks passed. Before/after probe inventories differ because review-driven alias and optional-mode cases were added.

All final boundary checks

Regression case	Result
services: foreign accounts/13788	PASS
services: foreign accounts/13788/transactions	PASS
services: foreign accounts/13788/transactions/amount/10.00	PASS
services: foreign transactions/14476	PASS
services: foreign customers/12656/accounts	PASS
services: encoded foreign identifier +13788	PASS
services: encoded foreign identifier %2b13788	PASS
services: encoded foreign identifier %313788	PASS
services: foreign debit rejected unchanged	PASS
services: own account works	PASS
services: unsupported authentication rejected	PASS
services_proxy: foreign accounts/13788	PASS
services_proxy: foreign accounts/13788/transactions	PASS
services_proxy: foreign accounts/13788/transactions/amount/10.00	PASS
services_proxy: foreign transactions/14476	PASS
services_proxy: foreign customers/12656/accounts	PASS
services_proxy: encoded foreign identifier +13788	PASS
services_proxy: encoded foreign identifier %2b13788	PASS
services_proxy: encoded foreign identifier %313788	PASS
services_proxy: foreign debit rejected unchanged	PASS
services_proxy: own account works	PASS
services_proxy: unsupported authentication rejected	PASS
invalid deposit abc rejected unchanged	PASS
invalid deposit 0 rejected unchanged	PASS
invalid deposit -1.00 rejected unchanged	PASS
invalid deposit 0.001 rejected unchanged	PASS
SOAP foreign denied	PASS
SOAP own allowed	PASS
Account XML root qualified	PASS
SOAP WSDL metadata accessible	PASS
Optional RESTJSON own account works	PASS
Optional RESTXML own account works	PASS

Regression case	Result
Optional SOAP own account works	PASS

Concurrent withdrawal results

api-seed-42: start=515.50 each=103.10 accepted=5 final=0.00 overdraft=false

api-seed-43: start=515.50 each=103.10 accepted=5 final=0.00 overdraft=false

The source account starts at 515.50 and each of ten synchronized requests asks for 103.10. At most five can succeed under the approved no-overdraft policy. Rejected withdrawals must not create ledger entries. The Java assertion compares the final balance with both accepted-request arithmetic and the signed ledger plus opening baseline. Full worker statuses/bodies/readiness/termination are retained in Allure attachments.

Six original defect reproduction checks

ID	HTTP	Original defect observed?	Before / after or XML root
API-01	403	NO	->
API-02	403	NO	{'victim': '525.5', 'requester': '515.5'} -> {'victim': '525.5', 'requester': '515.5'}
API-03	400	NO	515.5 -> 515.5
API-04	400	NO	515.5 -> 515.5
API-05	400	NO	415.5 -> 415.5
API-06	200	NO	-> {http://service.parabank.parasoft.com/}account

NOT REPRODUCED labels are supplementary: passing Java/33 boundary assertions establish ownership denial, unchanged balances/ledgers, input validation and XML qualification. The utility alone does not establish absence of every possible defect. Public historical findings were not reverified or withdrawn.

Complete final test inventory

ui-unit

Started 2026-10-07T15:44:37.3095813+05:30; finished 2026-10-07T15:44:50.4193833+05:30. Command from repository root: ui-automation/mvnw.cmd -q -f ui-automation/pom.xml clean test

Test case	Result	Seconds
CoreTest.envOverrides	PASS	0.193
CoreTest.missingSecret	PASS	0.004
CoreTest.rounding	PASS	0.010
CleanupTest.reverseOrderAndContinue	PASS	0.201
CartTotalTest.finalTotalWinsOverSubtotal	PASS	0.030
CartTotalTest.subtotalCannotStandInForMissingTotal	PASS	1.035

Evidence: ui-unit/execution.log, exit-code.txt, run-metadata.json, surefire-reports/ and allure-results/.

ui-chrome-isolation

Started 2026-10-07T15:44:51.0419672+05:30; finished 2026-10-07T15:45:12.8033933+05:30. Command from repository root: ui-automation/mvnw.cmd -q -f ui-automation/pom.xml clean test -Dgroups=browser

Test case	Result	Seconds
SessionTest.separateCookies	PASS	10.841

Evidence: ui-chrome-isolation/execution.log, exit-code.txt, run-metadata.json, surefire-reports/ and allure-results/.

ui-chrome

Started 2026-10-07T15:45:13.3705223+05:30; finished 2026-10-07T15:47:27.5684668+05:30. Command from repository root: ui-automation/mvnw.cmd -q -f ui-automation/pom.xml clean test -Pintegration

Test case	Result	Seconds
ScenarioTests.localizationAndCurrency	PASS	23.496
ScenarioTests.couponAndVoucher	PASS	27.309
ScenarioTests.orderLifecycleAndHistory	PASS	22.250
ScenarioTests.paginatedSearchFilterSort	PASS	5.540
ScenarioTests.requiredOptionsAndQuantity	PASS	1.075
ScenarioTests.returnStatusAndIsolation	PASS	21.040
ScenarioTests.stockAndCancellation	PASS	24.115

Evidence: ui-chrome/execution.log, exit-code.txt, run-metadata.json, surefire-reports/ and allure-results/.

ui-firefox-isolation

Started 2026-10-07T15:47:28.3514478+05:30; finished 2026-10-07T15:47:56.4411017+05:30. Command from repository root: ui-automation/mvnw.cmd -q -f ui-automation/pom.xml clean test -Dgroups=browser

Test case	Result	Seconds
SessionTest.separateCookies	PASS	16.587

Evidence: ui-firefox-isolation/execution.log, exit-code.txt, run-metadata.json, surefire-reports/ and allure-results/.

ui-firefox

Started 2026-10-07T15:47:56.9199220+05:30; finished 2026-10-07T15:51:27.9678653+05:30. Command from repository root: ui-automation/mvnw.cmd -q -f ui-automation/pom.xml clean test -Pintegration -Dthreads=1

Test case	Result	Seconds
ScenarioTests.couponAndVoucher	PASS	14.628
ScenarioTests.localizationAndCurrency	PASS	16.456
ScenarioTests.orderLifecycleAndHistory	PASS	16.863
ScenarioTests.paginatedSearchFilterSort	PASS	18.750

Test case	Result	Seconds
ScenarioTests.requiredOptionsAndQuantity	PASS	12.676
ScenarioTests.returnStatusAndIsolation	PASS	11.812
ScenarioTests.stockAndCancellation	PASS	16.461

Evidence: ui-firefox/execution.log, exit-code.txt, run-metadata.json, surefire-reports/ and allure-results/.

api-unit

Started 2026-10-07T15:59:38.0439065+05:30; finished 2026-10-07T15:59:49.4246646+05:30. Command from repository root: api-tests/mvnw.cmd -q -f api-tests/pom.xml clean test

Test case	Result	Seconds
ParityTest.detectsMissingFields	PASS	0.422
ParityTest.singletonCollectionHasParity	PASS	0.010
ParityTest.decimalParity	PASS	0.006
ParityTest.reorderedRecordsHaveParity	PASS	0.007
ParityTest.qualifiedFieldsHaveDecimalParity	PASS	0.007
ParityTest.externalEntityRejected	PASS	0.014
ParityTest.qualifiedEmptyCollectionHasParity	PASS	0.008
CoreTest.rounding	PASS	0.006
CoreTest.missingSecret	PASS	0.007
CoreTest.envOverrides	PASS	0.007
ResetClassificationTest.serverErrorIsNotReset	PASS	1.855
LedgerTest.unknownType	PASS	0.008
LedgerTest.creditsAndDebits	PASS	0.006

Evidence: api-unit/execution.log, exit-code.txt, run-metadata.json, surefire-reports/ and allure-results/.

api-seed-42

Started 2026-10-07T15:59:50.1008948+05:30; finished 2026-10-07T16:00:09.0836919+05:30. Command from repository root: api-tests/mvnw.cmd -q -f api-tests/pom.xml clean test -Pintegration -Dseed=42

Test case	Result	Seconds
ContractTests.deployedXmlSchema	PASS	0.933
ContractTests.jsonXmlParity	PASS	0.241
BankTests.movementAndLedger	PASS	1.235
BankTests.negativeWithdrawalCannotIncreaseBalance	PASS	0.088
BankTests.crossCustomerTransfer	PASS	0.328
BankTests.crossCustomerTransactionRead	PASS	0.180

Test case	Result	Seconds
BankTests.negativeDepositCannotReduceBalance	PASS	0.093
BankTests.transactionSearch	PASS	0.143
BankTests.concurrentWithdrawals	PASS	0.281
BankTests.invalidTransfers[insufficient, 999999.00](1)	PASS	0.391
BankTests.invalidTransfers[negative, -1.00](2)	PASS	0.369
BankTests.invalidTransfers[zero, 0](3)	PASS	0.339
BankTests.invalidTransfers[nonnumeric, abc](4)	PASS	0.285
BankTests.invalidTransfers[missing-target, 1.00](5)	PASS	0.134
BankTests.registrationAndIdentity	PASS	0.069

Evidence: api-seed-42/execution.log, exit-code.txt, run-metadata.json, surefire-reports/ and allure-results/.

api-seed-43

Started 2026-10-07T16:00:10.0189705+05:30; finished 2026-10-07T16:00:28.9590715+05:30. Command from repository root: api-tests/mvnw.cmd -q -f api-tests/pom.xml clean test -Pintegration -Dseed=43

Test case	Result	Seconds
BankTests.transactionSearch	PASS	0.272
BankTests.movementAndLedger	PASS	1.398
BankTests.negativeDepositCannotReduceBalance	PASS	0.102
BankTests.invalidTransfers[insufficient, 999999.00](1)	PASS	0.381
BankTests.invalidTransfers[negative, -1.00](2)	PASS	0.295
BankTests.invalidTransfers[zero, 0](3)	PASS	0.334
BankTests.invalidTransfers[nonnumeric, abc](4)	PASS	0.278
BankTests.invalidTransfers[missing-target, 1.00](5)	PASS	0.132
BankTests.concurrentWithdrawals	PASS	0.354
BankTests.crossCustomerTransfer	PASS	0.293
BankTests.negativeWithdrawalCannotIncreaseBalance	PASS	0.079
BankTests.registrationAndIdentity	PASS	0.069
BankTests.crossCustomerTransactionRead	PASS	0.128
ContractTests.deployedXmlSchema	PASS	0.694
ContractTests.jsonXmlParity	PASS	0.152

Evidence: api-seed-43/execution.log, exit-code.txt, run-metadata.json, surefire-reports/ and allure-results/.

api-loans

Started 2026-10-07T16:00:30.0814654+05:30; finished 2026-10-07T16:00:47.5394787+05:30. Command from repository root: api-tests/mvnw.cmd -q -f api-tests/pom.xml clean test -Dgroups=loans

Test case	Result	Seconds
LocalTests.loanRules	PASS	6.667

Evidence: api-loans/execution.log, exit-code.txt, run-metadata.json, surefire-reports/ and allure-results/.

api-reset

Started 2026-10-07T16:00:48.0636146+05:30; finished 2026-10-07T16:01:02.4017486+05:30. Command from repository root: api-tests/mvnw.cmd -q -f api-tests/pom.xml clean test -Dgroups=reset

Test case	Result	Seconds
LocalTests.midRunResetClassified	PASS	4.660

Evidence: api-reset/execution.log, exit-code.txt, run-metadata.json, surefire-reports/ and allure-results/.

Coverage limits and reproducibility

The loanRules method contains eight boundaries for a 100-unit loan and 20% local threshold: down payments 0.00,19.90,20.00,30.00,9999.00 and available funds 19.90,20.00,30.00. Settings restore in finally. The reset method deliberately removes local data, verifies fixture-loss classification, and initializes the bank in finally.

The 46-case risk plan is broader than automated coverage. No full upstream ParaBank unit/IT suite was executed; some upstream tests explicitly expect overdrafts and conflict with this approved local policy. The compiled overlay, assignment suites and real boundary regressions define the verified scope.

Local optional RESTJSON/RESTXML/SOAP own-account controls passed in the boundary probe; arbitrary remote endpoints were not certified. Credentials are not forwarded to arbitrary hosts. Administrative/reset endpoints remain available in this loopback-only demo. This is not a production security audit or deployment.

Native OpenCart percentage option pricing and customer-facing voucher remaining balance remain unavailable. A fixed reduction fixture and minimal QA Spanish labels are used. Categories/options/currencies/category associations use owned SQL; full admin-UI-only fixtures and complete localization certification are not claimed.

No load/stress, accessibility certification, email delivery, real settlement, Docker Grid, hosted CI or native JDK 17 run was performed. Browser CDP/Java deprecation warnings can appear; assertion/process results determine status. No automatic retries were used.

The original submission ZIP and previous reports were preserved. The new fixed submission ZIP excludes secrets, .git, target and caches. Synthetic bank data remains locally because per-customer deletion is unavailable. Application recreation resets ephemeral local data.

Top-level Surefire TEST*.xml determines counts; duplicate JUnit exports and lowercase testng-results.xml are excluded. results.json and test-case-results.csv contain final case details. Diagnostics remain separate and visible. Raw_Run_Evidence.zip includes metadata, probes, logs, HTML/XML reports and Allure attachments.

To reproduce: follow api-tests/patch/README.md in the fixed submission. Start the bank with both docker-compose.yml and docker-compose.patched.yml and --build --wait. Set API_BASE=http://127.0.0.1:8081/parabank; set JAVA_HOME and dot-source scripts/configure.ps1. Use the per-run commands listed above, UI_BROWSER=chrome/firefox and -Dthreads=1 for Firefox. Run scripts/test_summary.py and scripts/verify_bank_patch.py separately. See source-file-checksums.json and patched-image-id.txt for this exact delivered snapshot.